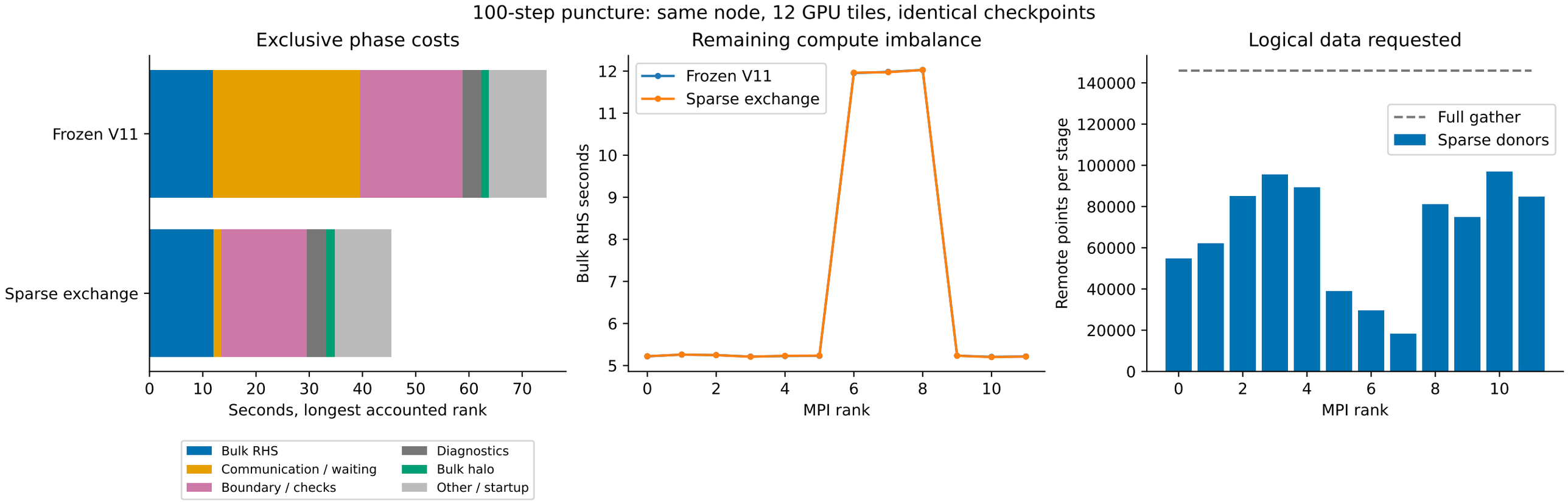
100-step puncture: same node, 12 GPU tiles, identical checkpoints
Exclusive phase costs
Frozen V11
Sparse exchange
0
10
20
30
40
50
60
70
Seconds, longest accounted rank
Bulk RHS
Communication / waiting
Boundary / checks
Diagnostics
Bulk halo
Other / startup
Remaining compute imbalance
Frozen V11
Sparse exchange
Bulk RHS seconds
5
6
7
8
9
10
11
12
0
2
4
6
8
10
MPI rank
Logical data requested
Full gather
Sparse donors
Remote points per stage
0
20000
40000
60000
80000
100000
120000
140000
0
2
4
6
8
10
MPI rank